INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

CAKES, SHELF STABLE, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers shelf stable cakes (oven baked process for low water activity batters) in flexible pouches for use as a component in the Meal, Ready-To-Eat, Individual.

1.2 Classification. The product shall be of the following types as specified (see 6.1).

Types

I	-	Vanilla pound cake
II	-	Lemon pound cake
III	-	Orange pound cake
IV	-	Pineapple pound cake
V	-	Chocolate mint pound cake with chocolate drops
VI	-	Fudge brownie with chocolate drops

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8920

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are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-636	-	Boxes, Shipping, Fiberboard

STANDARDS

FEDERAL

FED-STD-595	-	Colors Used in Government Procurement

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-900	-	Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

13-1-0185 - Cakes, Shelf Stable, For Meal, Ready-To-Eat, Level C Shipping Container and S-Shaped Liner, Figure 1

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UXD, Natick, MA 01760-5017.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing the Inspection of Egg and Egg Products (7 CFR Part 59)

(Copies are available from the Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

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AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC International

(Application for copies should be addressed to the AOAC International, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Flour, cake. Flour shall be soft wheat cake flour, bleached, enriched or unenriched. If unenriched is used, equivalent enrichment shall be finely incorporated and dispersed by the manufacturer at the time of production of the finished product. The flour shall comply with MIL-STD-900.

3.2.2 Sugar, white (for bakery). Sugars shall be white, refined cane, or beet sugars or a combination thereof. The particle size shall be a fine grind suited for use in bakery products and shall comply with MIL-STD-900.

3.2.3 Starch, instant, granular. The starch shall be a white to off-white odorless finely ground powder. A five percent starch solution hydrated in cold or ambient temperature water shall be a smooth, non-grainy paste that develops to a full, spoonable consistency in five minutes. The starch shall have a maximum moisture content of 8.0 percent and a pH of 5.0 - 7.0. The prepared starch shall display good clarity and a thick, smooth, glossy consistency with good cold storage stability. It shall be bland with essentially no cereal or starch taste (see 6.3.1).

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3.2.4 Maltodextrin. Maltodextrin shall be a white, free flowing powder with a dextrose equivalent of ten. It shall have a maximum moisture of 6 percent, a pH of 4.0 - 4.7 and have 35 - 40 percent solids.

3.2.5 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.6 Baking powder. Baking powder shall be of the type known as "Double Acting". The baking powder shall be prepared from highly purified food grade materials. The baking powder shall be a white, free-flowing mixture of sodium bicarbonate, corn starch, and one or more of the acid-reacting substances monocalcium phosphate, sodium acid pyrophosphate, and sodium aluminum phosphate. The baking powder shall have an available carbon dioxide level of not less than 13 percent, by weight.

3.2.7 Xanthan gum. Xanthan gum shall comply with the Food Chemicals Codex.

3.2.8 Guar gum. Guar gum shall comply with the Food Chemicals Codex.

3.2.9 Potassium sorbate. Potassium sorbate shall comply with the Food Chemicals Codex.

3.2.10 Shortening, high ratio. High ratio shortening shall be a highly stable emulsified, white, plastic shortening made from hydrogenated soybean or cottonseed oil or a blend of these oils. The high ratio shortening shall have a maximum lovibond color of 2.0R; a Wiley melting point range of 111 to 119°F; maximum free fatty acids of 0.1 percent and an alpha monoglycerol content of 3.0 to 3.6 percent. It shall have a stability of not less than 100 hours (A.O.M.) (see 6.3.2).

3.2.11 Eggs, whole, frozen. Whole eggs shall be frozen and shall have been processed and labeled in accordance with the Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59). The whole eggs shall be egg whites and egg yolks in their natural proportions as broken directly from the shell eggs, as evidenced by a USDA Egg Products Inspection Certificate. The frozen whole eggs shall be USDA certified salmonella-free and shall conform to the requirements for Frozen Eggs (21 CFR Part 160.110). The USDA certificate shall also state the date of pasteurization. The USDA certificate shall accompany egg products shipped between plants. Frozen whole eggs shall be held at 10°F or lower and used within 120 days from the date of pasteurization. The whole eggs shall be free from off-odors and off-flavors, free from foreign odors and flavors such as metallic, musty, sour, or fruity, and free from foreign colors and materials.

3.2.12 Eggs, whole, dried. The dried whole eggs shall be processed and labeled in accordance with the USDA Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59). The dried whole eggs shall be USDA certified salmonella free and shall conform to the FDA Standards of Identity

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for Dried Eggs (21 CFR Part 160.105). The dried whole eggs shall not have been held in storage for more than 90 days prior to use in product preparation (see 3.3). The USDA certificate shall accompany egg products shipped between plants.

3.2.13 Water. Water used for formulation and washing shall conform to the National Primary Drinking Water Regulations.

3.2.14 Glycerol. Glycerol shall be clear, colorless, and odorless and shall comply with the Food Chemicals Codex.

3.2.15 Cocoa (for type V cake). Cocoa shall be medium fat (10 to 12 percent cocoa fat), non-dutched type cocoa. The cocoa shall conform to the Definition and Standards for cocoa products under the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.2.16 Flavoring, cream. The flavoring shall be a liquid or a powder and shall have a typical cream flavor and aroma.

3.2.17 Flavoring, vanilla. Vanilla flavoring shall be natural or artificial double strength flavoring (see 6.3.3).

3.2.18 Lemon oil. Lemon oil shall be of a two fold concentration level or equivalent concentration if other than two-fold and shall be cold pressed from fresh, sound, mature lemons of one or more of the high acid varieties. The lemon oil shall comply with the Food Chemicals Codex (see 6.3.4).

3.2.19 Orange oil. Orange oil shall be a two-fold concentration level or equivalent concentration if other than two-fold and shall be cold pressed from Valencia oranges. The orange oil shall comply with the Food Chemicals Codex (see 6.3.5).

3.2.20 Flavoring, imitation pineapple. The imitation flavoring shall have a typical pineapple flavor, color and odor (see 6.3.6).

3.2.21 Flavoring, chocolate mint. The flavoring shall be a liquid or a powder and shall have a typical chocolate mint flavor and aroma (see 6.3.7).

3.2.22 Chocolate drops. Chocolate drops shall be semi-sweet chocolate having a count per pound of not less than 2200 nor more than 4000. They shall be free of dextrose and contain not less than 28 percent fat.

3.2.23 Yellow lake dispersion. Yellow lake dispersion shall be an aluminum salt of FD & C water-soluble dyes extended on a substratum of aluminum. It shall be a certified FD & C yellow #5.

3.2.24 Orange lake dispersion. Orange lake dispersion shall be an aluminum salt of FD & C water-soluble dyes extended on a substratum of aluminum. It shall be a certified mixture of FD & C yellow #5 and FD & C yellow #6.

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3.2.25 Citric acid, anhydrous. Anhydrous citric acid shall comply with the Food Chemicals Codex.

3.2.26 Cocoa (for type VI cake). The cocoa shall have 11 to 15 percent fat, a maximum moisture of 5.0 percent, and shall have a pH of 5.3 to 5.9. The cocoa shall conform to the Definition and Standards for cocoa products under the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder (see 6.3.8).

3.2.27 Shortening, vegetable. Vegetable shortening shall be refined, hydrogenated, deodorized cottonseed, peanut, soybean, or palm oils, and may contain antifoaming agents and antioxidants (active ingredients BHA, BHT, TBHQ, or propyl gallate) in amounts permitted by the FDA. The vegetable shortening shall possess a uniform plastic texture and shall have a stability of not less than 100 hours (A.O.M.).

3.2.28 Emulsifier. The emulsifier shall be sucrose fatty acid esters, complying with the Code of Federal Regulations (21 CFR, Part 172.859) and shall be limited to sucrose ester stearate and having an HLB number of approximately 16.

3.2.29 Sodium steroyl-2-lactylate. Sodium steroyl-2-lactylate shall be a free flowing powder and shall comply with the Food Chemicals Codex.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Batter formulation (percent by weight).

Ingredient	Type I	Type II	Type III	Type IV	Type V	Type VI
Flour, cake	24.12	24.06	23.98	24.00	16.33	13.79
Sugar, granulated	27.69	27.65	27.65	27.65	24.91	24.25
Starch, instant, granular	1.00	1.00	1.00	1.00	1.00	1.00
Maltodextrin	2.50	2.50	2.50	2.50	2.50	-----
Salt	0.34	0.34	0.34	0.34	0.30	0.28
Baking powder	0.15	0.15	0.15	0.15	0.13	-----
Xanthan gum	0.03	0.03	0.03	0.03	0.03	0.11
Guar gum	0.03	0.03	0.03	0.03	0.03	0.11
Potassium sorbate	0.05	0.05	0.05	0.05	0.05	0.05
Shortening, high ratio	16.25	16.25	16.25	16.25	13.73	-----
Eggs 1/	17.28	17.28	17.28	17.28	15.57	11.26
Water	6.91	6.91	6.91	6.91	6.23	13.57
Glycerol	3.46	3.46	3.46	3.46	3.12	2.13
Cocoa (see 3.2.15)	-----	-----	-----	-----	6.00	-----
Flavoring, cream	0.04	0.04	0.04	0.04	0.04	-----
Flavoring, vanilla	0.15	0.11	0.11	0.11	0.10	0.94

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Lemon oil	-----	0.05	-----	-----	-----	-----
Orange oil	-----	-----	0.07	-----	-----	-----
Flavoring, pineapple	-----	-----	-----	0.07	-----	-----
Flavoring, chocolate mint	-----	-----	-----	-----	0.03	-----
Chocolate drops	-----	-----	-----	-----	9.90	10.79
Yellow lake dispersion	-----	0.02	-----	0.01	-----	-----
Orange lake dispersion	-----	-----	0.08	-----	-----	-----
Citric acid, anhydrous	-----	0.07	0.07	0.12	-----	-----
Cocoa (see 3.2.26)	-----	-----	-----	-----	-----	6.46
Shortening, vegetable	-----	-----	-----	-----	-----	15.02
Emulsifier	-----	-----	-----	-----	-----	0.12
Sodium steroyl-2-lactylate	-----	-----	-----	-----	-----	0.12
	100.00	100.00	100.00	100.00	100.00	100.00

1/ a. Frozen whole eggs shall be thawed and held at a temperature of 40°F or lower until defrosted and shall be held for not more than 24 hours after defrosting prior to use in the formula. In no case shall the defrosting period exceed 5 days.

b. Dry whole eggs may be substituted for whole eggs (liquid basis) by following the manufacturers recommended rehydration and mixing procedure and shall have no less than the equivalent amount of whole egg solids as the liquid basis.

3.3.2 Preparation.

3.3.2.1 Preparation of type I-V cakes.

a. The cake batter shall be prepared in accordance with good commercial practice to meet the finished product requirements. 1/

b. When preparing type V cake, the chocolate drops shall be added in the final mixing step and minimally mixed.

1/ Batter specific gravity in grams per cubic centimeter of batter shall be not less than 0.86 or more than 0.97 (see 4.5.2.1).

3.3.2.2 Preparation of type VI cake.

a. Mix together all dry ingredients except sugar, sodium steroyl-2-lactylate and chocolate drops. Set aside.

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b. Add sodium steroyl-2-lactylate to the shortening and sugar. Cream the mixture until light and fluffy.

c. Slowly mix the dry ingredients with the creamed sugar and shortening. After all dry ingredients have been added the mixture shall appear uniform.

d. The liquid ingredients may be combined or added separately. If combined, the liquid ingredients shall not be frothy. The liquid ingredients shall be slowly mixed into the batter mixture.

e. Blend sufficiently to obtain brownie meeting finished product requirements. 1/

f. Add the chocolate drops to the batter and mix minimally, but uniformly.

1/ Batter specific gravity in grams per cubic centimeter of batter shall be not less than 0.90 or more than 0.98 (see 4.5.2.1).

NOTE: The above preparation procedures were used in the development of the cakes. Alternative procedures may be used provided finished product requirements are met.

3.3.3 Baking. The cake batter shall be poured into baking molds to yield rectangular baked cakes which dimensions measure not more than 4.75 inches in length, not more than 3.25 inches in width, and not more than 1.125 inches in height with flat top and bottom surface, after baking. The cakes shall be fully baked until the exterior is a uniform, typical cake surface color and then cooled to a temperature range of 100 $\pm$ 20°F prior to filling. The time from baking to filling shall be not longer than 2 hours.

NOTE: If cake rises excessively, amount of leavening may be reduced to adjust cake volume.

3.4 Pouch filling and sealing. One unit of cake and one package of an FDA approved oxygen scavenger shall be placed into a pouch in accordance with 5.1.1.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material, such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

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d. The average net weight of type I-V cakes shall not be less than 2.50 ounces.

e. No individual pouch of type I-V cakes shall have a net weight less than 2.25 ounces.

f. The average net weight of type VI product shall be not less than 3.00 ounces.

g. No individual pouch of type VI product shall contain less than 2.60 ounces.

h. The oxygen content in an individual pouch shall not be more than 0.30 percent.

i. Each pouch shall contain one intact unit of cake and one intact packet of oxygen scavenger.

j. The water activity for an individual pouch shall be not more than 0.85 when measured at 25°C.

k. The units of cake shall have a rectangular shape, proper dimensions (see 3.3.3) and appearance.

l. The cake crumb and surface shall have a color typical of the product.

m. The cake shall have a flavor typical of its type.

n. The texture of the cake shall not be excessively dry and crumbly or moist and gummy.

o. The cake shall show no evidence of dense crumb compression streaks.

3.5.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.6 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision and inspection necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5.1.2 Pouch material certification. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements.

Material requirement	Requirement paragraph	Test procedure
Thickness of films	5.1.1.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Aluminum thickness	5.1.1.1	As specified in QQ-A-1876.
Laminated material identification and construction	5.1.1.1	Laboratory evaluation
Color of laminated material	5.1.1.1	Visual evaluation

4.5.1.3 Unfilled preformed pouch certification. A certificate of compliance may be accepted as evidence that unfilled pouches conform to the requirements specified in 5.1.1. When deemed necessary by the USDA, testing of the unfilled preformed pouches for seal strength shall be as specified in 4.5.1.3.1.

4.5.1.3.1 Unfilled preformed pouch seal strength testing. The unfilled pouches shall be tested for seal strength in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced two inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in pouches. The sample size shall be the number of pouches indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement or any seal failing to meet the average seal strength requirement specified in 5.1.1.1.1 shall be classified as a major defect and be cause for rejection of the lot.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

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4.5.2.1 In-process examination of cake specific gravity. After establishing and standardizing mixing conditions that result in the required specific gravity, cake batter shall be examined once per batch as follows for conformance to the specific gravity requirements in 3.3.2.1 and 3.3.2.2:

If not known, a bakers batter specific gravity determination cup or reasonable substitute shall be standardized as follows: Obtain tare weight of the cup. Fill cup with water. Level top of water with a straight edged spatula. Subtract the tare weight of the cup from the total weight of the cup and water to obtain the net weight (in grams) of water. This is a very close approximation of the volume of water in cubic centimeters. This procedure need only be done once if the same cup is always used for specific gravity determination.

The dry cup used for specific gravity determination shall be uniformly filled with batter so as to prevent entrapment of air. The top of the batter shall then be leveled with the lip of the cup using a spatula or other appropriate straight edge. The net weight of the batter shall be obtained by subtracting the total weight from the tare weight of the cup. The batter specific gravity = batter fill weight/net weight of an equal volume of water.

Failure to comply with specific gravity requirements shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

TABLE I. Filled and sealed pouch defects 1/

Category		Defect
Major	Minor	
101		Tear, hole, or open seal
102		Pouch has foreign odor
103		Not clean 2/
104		Evidence of delamination 3/
105		Seal width less than 1/16 inch 4/
	201	Evidence of delamination 3/
	202	Required labeling missing, incorrect, illegible, or smudged
	203	Tear notch missing or does not facilitate easy opening
	204	Closure or top seal extends into or below tear notch

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1/ Any evidence of insect or rodent infestation shall be cause for rejection of the lot.

2/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal fillings, etc.) or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean, dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

3/ Major - Evidence of outer ply delamination such that aluminum foil in the pouch body is exposed or evidence of two ply delamination such that the clear plastic food contact layer is exposed. Delamination of the outer ply in the pouch seal area that can be propagated to expose aluminum foil at the food product edge of the pouch after manual flexing of the delaminated area; To flex, the delaminated area shall be held between the thumb and forefingers touching each other. The delaminated area shall then be rapidly flexed 10 times by rotating both hands in alternating clockwise-counter clockwise directions. Care shall be exercised when flexing delaminated areas near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be classified as a major defect. Additionally, spot delamination of the outer ply in the body of the pouch that is able to be propagated beyond its initial borders shall be classified as a major defect. To determine if the delaminated area is a defect, use the following procedure: Mark the outside edges of the delaminated area using a bold permanent marking pen. Open the pouch and remove the contents. Cut the pouch transversely not closer than 1/4 inch ($\pm$ 1/16 inch) from the delaminated area. The pouch shall be flexed in the area in question using the procedure described above. Any propagation of the delaminated area, as evidenced by the delaminated area exceeding the limits of the outlined borders, shall be classified as a major defect.

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Minor - Evidence of minor delamination in the seal area or isolated spots of delamination in the body of the pouch that does not propagate when flexed as described above.

4/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide, from side seal to side seal that produces a hermetically sealed pouch.

4.5.4 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 2.25 ounces for types I-V cakes or 2.60 ounces for type VI cake shall be classified as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.01 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 2.50 ounces for types I-V cakes or 3.00 ounces for type VI cake.

4.5.5 Product inspection. The filled and sealed sample pouches shall be conditioned to 70 to 80°F, opened, and inspected for defects indicated in table II. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE II. Product defects 1/ 2/

Category		Defect
Major	Minor	
101		Oxygen content in pouch exceeding 0.30 percent 3/
102		Pouch does not contain one intact unit of cake and one intact packet of oxygen scavenger
103		Texture of cake is excessively dry, crumbly, or excessively moist and gummy
104		Cake shows evidence of dense crumb compression streaks 4/
105		Cake flavor not typical for its type (see 1.2)
106		Tear, hole, or open seal in oxygen scavenger packet
	201	Crust color not as specified
	202	Pouch contains a unit of cake that does not have the required shape, appearance, and dimensions (see 3.3.3)
	203	Crumb color not as specified

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1/ Presence of foreign material (for example, glass, dirt, insect part, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, moldy, musty), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.5.1).

3/ Filled and sealed pouches shall be tested for oxygen content in accordance with any USDA approved test method. Testing for oxygen content shall be accomplished only after filled and sealed pouches have been allowed to equilibrate at room temperature for not less than 36 hours from time of sealing.

4/ To inspect for this defect, cut cake units in half along the length from top to bottom.

4.5.6 Pouch closure seal strength testing. The filled and sealed pouches shall be tested for closure seal strength in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply tensile load to the specimen by movement of the upper or lower clamp may be used. The test specimens shall be cut to a length suitable for proper mounting. Three adjacent specimens, 1/2 or 1 inch wide shall be cut from the closure seal of each pouch in the sample. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from that seal. The results shall be reported to the nearest 0.1 pound per inch of width. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The sample size shall be the number of pouches indicated by inspection level S-1. Any individual test specimen or average closure seal strength failing to meet the requirements of 5.1.1.1.2 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Water activity testing. Eight filled and sealed pouches shall be selected at random from the lot regardless of lot size. Water activity shall be determined not less than 4 days but not more than 14 days after baking to allow moisture equilibration in the product. The pouched product shall be individually tested for water activity in accordance with the Official Methods of Analysis of the AOAC method 978.18, using an electric hygrometer system self temperature controlled (at 25°C) or an equivalent instrument. The sample unit shall be a specimen from the center of the cake. The results of each Aw (water activity) determination shall be reported to the nearest 0.01. Any test result failing to conform to the requirements in 3.5 shall be cause for rejection of the lot.

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4.5.8 Shipping container examination. Shipping containers shall be examined in accordance with the appendix of PPP-B-636. In addition, the following defects shall be included in the table of examination:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Interior packing with fiberboard liner not as specified.
S-shaped liner missing or not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement of pouches not as specified.

5. PACKAGING

5.1 Packaging. Preservation shall be level A.

5.1.1 Level A. One cake of one type only shall be unit packed in a barrier bag (see 5.1.1.1) with one oxygen scavenger packet as specified in 5.1.1.2.

5.1.1.1 Preformed pouch. The preformed pouch shall be fabricated from 0.002 inch thick ionomer or polyethylene film laminated or extrusion coated to 0.00035 inch thick aluminum foil which is then laminated to 0.0005 inch thick polyester. The three plies shall be laminated with the polyester on the exterior of the pouch. The complete exterior surface of the pouch shall be uniformly colored in the range of 34079 through 34087 or 24052 through 24087 or 30045 through 30118 (excluding 30109) or 10045 of FED-STD-595. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches.

5.1.1.1.1 Pouch construction. The preformed pouch shall be a flat style pouch having inside dimensions of 5 inches wide by 6-5/8 inches long (± 1/8 inch). The pouch shall be made by heat sealing three edges with 3/8 inch (± 1/8 inch) wide seals. The heat seals shall be made in a manner that will assure hermetic seals. The side and bottom seals shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch when tested as specified in 4.5.1.3.1. A V-shaped or C-shaped (half round) tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top or bottom edge of the pouch, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch may be provided with an extended or foldover lip, extended not more than 1/8 inch (± 1/16 inch) to facilitate opening and filling. Tear notch location shall be measured from the top or bottom of the pouch, excluding the extended or foldover lip.

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baked, processed, and 3.6 Plant qualification. The product shall be pre

5.1.1.1.2 Pouch filling and sealing. One baked cake and one package of oxygen scavenger shall be placed into the pouch in a manner so as to avoid contamination of the closure seal area. The filled pouch shall be closed with a continuous heat seal not less than 1/4 inch wide. If thermal impulse or combination (heated curved bar with thermal impulse) sealing is used, any seal width from 1/8 to 7/16 inch will be acceptable. The average seal strength shall be not less than 6 pounds per linear inch, and no individual test specimen shall be less than 5 pounds when tested as specified in 4.5.6.

5.1.1.2 Oxygen scavenger packet. The oxygen scavenger shall be FDA approved as suitable for use with food. The oxygen scavenger packet shall be resistant to the migration of oil, moisture and scavenger components (see 3.5, 4.5.5 and 6.3.9).

5.2 Packing. Packing shall be level C.

5.2.1 Level C. One hundred cakes of one type only, packaged as specified in 5.1, shall be packed into a fiberboard shipping container constructed and closed in accordance with style RSC-L, class domestic, variety SW, grade 200 of PPP-B-636. The cakes shall be arranged on edge, twenty five in length, two in width and two in depth. The inside dimensions of the box shall be as shown in figure 1. An S-shaped liner/partition, constructed as shown in figure 1 shall be placed inside the box dividing the contents into two sections of fifty cakes each.

5.3 Labeling and marking.

5.3.1 Pouches. Each pouch shall be clearly printed with permanent ink in large letters of black, purple, or similar dark contrasting color with the following information:

(Flavor, as applicable) pound cake, shelf stable 1/
(name and address of producer)
Date of pack (expressed as a four digit Julian code)

1/ For type VI product, label shall read "Fudge brownie, shelf stable". The name and address of producer, and date of pack shall also be included.

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

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b. Type of product required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issues of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.5.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Ingredient information.

6.3.1 Starch, instant, granular. It has been found that the product (Mirasperse) manufactured by A.E. Staley, Manufacturing Co., 2200 Eldorado St., Decatur, IL 62525 meets the requirements of 3.2.3 and performs satisfactorily in this product.

6.3.2 Shortening, high ratio. It has been found that product manufactured by Van Den Bergh Food Ingredients Group, P.O. Box 796, Joliet, IL 60434-0796 meets the requirements of 3.2.10 and performs satisfactorily in this product.

6.3.3 Flavoring, vanilla. It has been found that product Nos. Natural H-9094 and Artificial 262003 manufactured by Haarman and Reimer Corp., Box 175, Springfield, NJ 07081 meet the requirements of 3.2.17 and perform satisfactorily in this product.

6.3.4 Lemon oil. It has been found that product No. 055905 manufactured by Felton Worldwide, 599 Johnson Ave., Brooklyn, NY 11237 meets the requirements of 3.2.18 and performs satisfactorily in this product.

6.3.5 Orange oil. It has been found that product No. 057422 manufactured by Felton Worldwide, 599 Johnson Ave., Brooklyn, NY 11237 meets the requirements of 3.2.19 and performs satisfactorily in this product.

6.3.6 Flavoring, imitation pineapple. It has been found that product No. US64803 manufactured by Haarman and Reimer Corp., Box 175, Springfield, NJ 07081 meets the requirement in 3.2.20 and performs satisfactorily in this product.

6.3.7 Flavoring, chocolate mint. It has been found that product Nos. Natural 7996 and Artificial 261952 manufactured by Haarman and Reimer Corp., Box 175, Springfield, NJ 07081 meet the requirements of 3.2.21 and perform satisfactorily in this product.

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acceptable quality level (AQL), expressed in terms of ... shall be 3.5 and the ...

6.3.8 Cocoa (for type VI). It has been found that Hershey's American Process Medium Fat Cocoa, product number 05100, meets the requirements of 3.2.26 and performs satisfactorily in this product.

6.3.9 Oxygen scavenger. It has been found that product manufactured by Multiform Desiccants, Inc., 960 Busti at Niagara, Buffalo, NY 14213 meets the requirements of 3.5, 4.5.5 and 5.1.1.2 and performs satisfactorily in this product.

6.4 Subject term (key word) listing.

Combat field feeding
Operational ration
Pouch food
Shelf stable

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 35

Review activities:

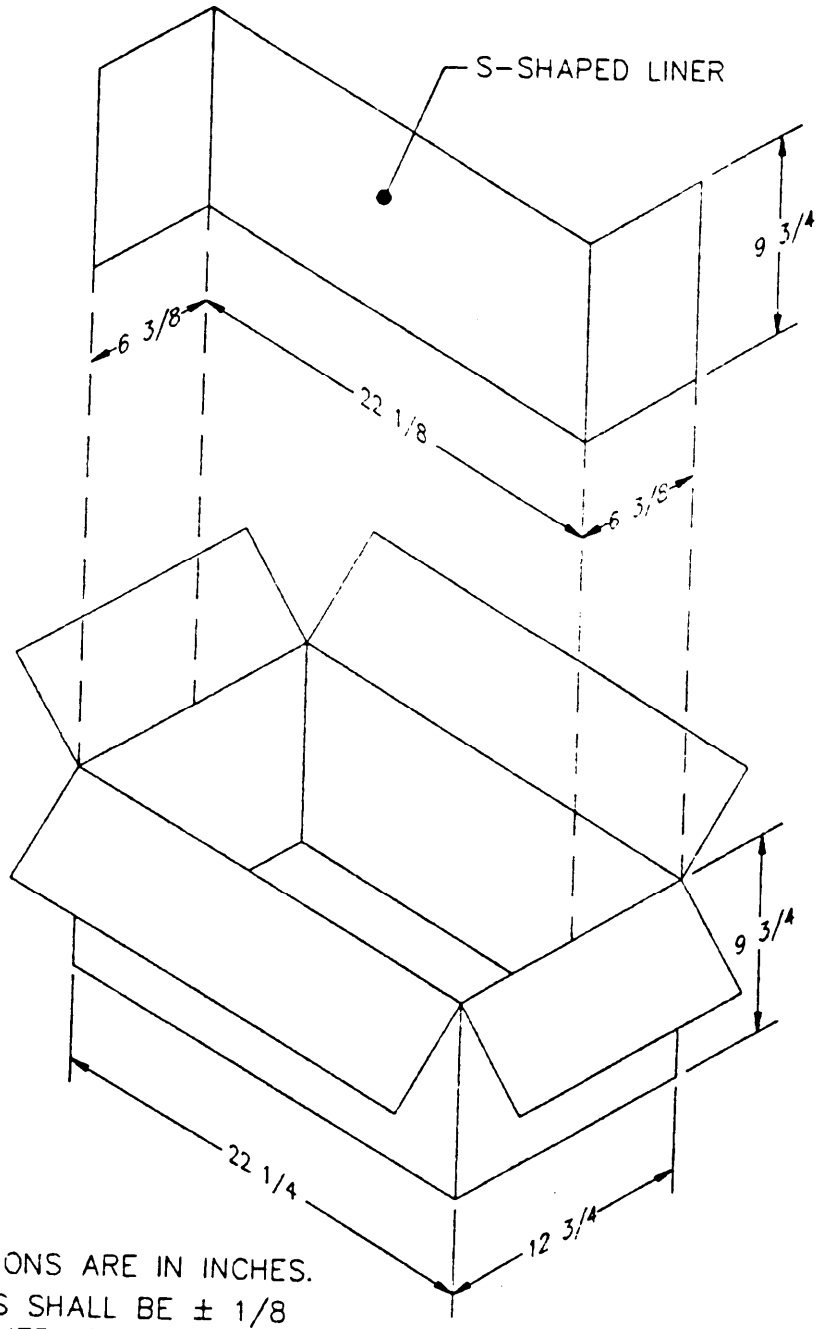
Army - MD, QM
Navy - MC
DLA - SS

Preparing activity:

Army - GL

(Project 8920-0558)

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NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. TOLERANCES SHALL BE ± 1/8 UNLESS OTHERWISE NOTED.

FIGURE 1. LEVEL C SHIPPING CONTAINER & S-SHAPED LINER.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-44071D	1994 March 31

3. DOCUMENT TITLE
CAKES, SHELF-STABLE, FOR MEAL, READY-TO-EAT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON/DSN 256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WRE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP4, JUL 91 *Previous editions are obsolete.* 198/29

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